

File Permissions

```
himasree@LAPTOP-41COT5AA:~$ cd ~/OSLab
himasree@LAPTOP-41COT5AA:~/OSLab$ nano permissions.c
himasree@LAPTOP-41COT5AA:~/OSLab$ gcc permission.c -o permission
cc1: fatal error: permission.c: No such file or directory
compilation terminated.
himasree@LAPTOP-41COT5AA:~/OSLab$ gcc permissions.c -o permissions
himasree@LAPTOP-41COT5AA:~/OSLab$ ./permissions
File created successfully.
himasree@LAPTOP-41COT5AA:~/OSLab$ ls -l sample.txt
-rwxrwxrwx 1 himasree himasree 28 Aug 22 03:03 sample.txt
himasree@LAPTOP-41COT5AA:~/OSLab$ |
```

- 1) Give Read and Write only to the owner

```
himasree@LAPTOP-41COT5AA:~/OSLab$ chmod 600 sample.txt
himasree@LAPTOP-41COT5AA:~/OSLab$ ls -l sample.txt
-rw----- 1 himasree himasree 28 Aug 22 03:03 sample.txt
himasree@LAPTOP-41COT5AA:~/OSLab$ |
```

- 2) Give Read and Write to the owner and group

```
himasree@LAPTOP-41COT5AA:~/OSLab$ chmod 660 sample.txt
himasree@LAPTOP-41COT5AA:~/OSLab$ ls -l sample.txt
-rw-rw---- 1 himasree himasree 28 Aug 22 03:03 sample.txt
himasree@LAPTOP-41COT5AA:~/OSLab$ |
```

- 3) Only owner can read and write

```
himasree@LAPTOP-41COT5AA:~/OSLab$ chmod 600 sample.txt
himasree@LAPTOP-41COT5AA:~/OSLab$ ls -l sample.txt
-rw----- 1 himasree himasree 28 Aug 22 03:03 sample.txt
```

- 4.)Owner read/write, everyone else read-only

```
himasree@LAPTOP-41COT5AA:~/OSLab$ chmod 644 sample.txt
himasree@LAPTOP-41COT5AA:~/OSLab$ ls -l sample.txt
-rw-r--r-- 1 himasree himasree 28 Aug 22 03:03 sample.txt
```

5.) Owner can do everything, others can only read and execute

```
himasree@LAPTOP-41COT5AA:~/OSLab$ chmod 755 sample.txt
himasree@LAPTOP-41COT5AA:~/OSLab$ ls -l sample.txt
-rwxr-xr-x 1 himasree himasree 28 Aug 22 03:03 sample.txt
himasree@LAPTOP-41COT5AA:~/OSLab$ |
```

6.) Owner and group can read/write

```
himasree@LAPTOP-41COT5AA:~/OSLab$ chmod 660 sample.txt
himasree@LAPTOP-41COT5AA:~/OSLab$ ls -l sample.txt
-rw-rw---- 1 himasree himasree 28 Aug 22 03:03 sample.txt
```

7.) Only owner can execute

```
himasree@LAPTOP-41COT5AA:~/OSLab$ chmod 700 sample.txt
himasree@LAPTOP-41COT5AA:~/OSLab$ ls -l sample.txt
-rwx----- 1 himasree himasree 28 Aug 22 03:03 sample.txt
```

8) Owner: read + write; Group: nothing; Others: nothing → 600

9) Owner: read + write; Group: read; Others: read → 644

10) Owner: read + write + execute; Group: read + execute; Others: read + execute → 755

11) Owner: read + write; Group: read + write; Others: nothing → 660

12) Everyone: read only → 444

```
himasree@LAPTOP-41COT5AA:~/OSLab$ cat permissions.c
#include <stdio.h>
int main()
{
    FILE *fp;
    fp = fopen("sample.txt", "w");
    if (fp == NULL)
    {
        printf("Cannot open file\n");
        return 1;
    }
    fprintf(fp, "Hello, this is a test file.\n");
    fclose(fp);
    printf("File created successfully.\n");
    return 0;
}
```

